# Transfer task: plan and begin

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Organize the given evidence and begin an independent position prediction.

**SUCCESS CRITERIA - I can...**

1. Organize the task data with fixed axes and matching units.
2. Calculate central and bounded final coordinates.
3. Save an explicit plan for the graph and explanation.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “component” and “bound” to explain your model or result. Sentence starters are optional.

**component**: the part of a vector along one chosen axis

**bound**: an upper or lower limit in a model

**assumption**: a condition accepted for using a model

## 01 Bring back an earlier idea

A 3 kg lamp hangs at rest from a vertical rope.
**Use** g=9.8.
**Find** tension. Then find tension if it accelerates upward at 2 m/s².

**Word help:** tension = the pulling force through a stretched rope or cable

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $v_x = v \cos\theta$; $v_y = v \sin\theta$; $x_f = x_i + v_x t$
$y_f = y_i + v_y t$.
Fictional fixed-Earth model: $x_i = -1.2\times10^8$ km; $y_i = 0$; $v = 8500$ m/s; $\theta = 15°$ from +x toward +y; $t = 1.6\times10^7$ s. Convert units.

**Worked reasoning:** A toy model uses $x_i$=0 m and constant v=2 m/s for 5 s, giving x=10 m. This is a conditional prediction, not proof that the real object follows the model.

**Your turn: Draw** the predicted path and mark one observation you would collect to test the constant-velocity assumption.

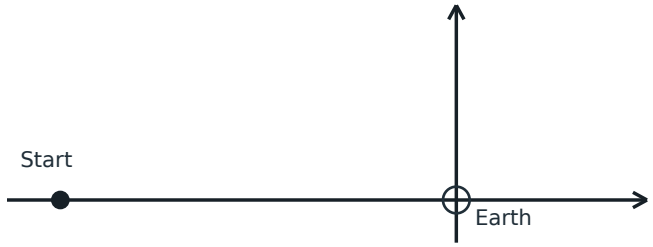
**Sentence starter:** I predict ... . My model supports this because ... .

**Marzano check - circle:** ① ② ③

## 03 Plan and represent

**Draw** the coordinate convention and write an ordered plan for units, velocity components, endpoint, bounds, graph and public explanation.

**Sentence starter:** My design shows ... . It meets the requirement by ... .




The small starting dot is the object; the labeled circle marks Earth at the origin. Not to scale.

**Marzano check - circle:** ① ② ③

## 04 Central prediction

Resolve velocity and calculate final x, y in km. Distinguish final position from displacement and separation. Keep unrounded components until the end.

**Word help:** resolve = separate a vector into components

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Propagate bounded inputs

Initial x bound±50, 000 km; speed bound±200 m/s at fixed 15°. Initial x and speed vary independently; angle, $y_i$, time exact.
**Find** coordinate half-widths and intervals.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** 1 2 3

## 06 Choose a method

A puck follows a circle at a steady 7 m/s. A student calls it equilibrium.
**Draw** velocity at two points and correct the claim.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .

**Marzano check - circle:** 1 2 3

## 07 Checkpoint

Prepare central/lower/upper x(t) values at t 0 and final time for tomorrow’s graph. Write one assumption, one excluded physical effect, and the next unfinished step.

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

A model predicts x=10 m while the measurement is 8.0±0.2 m. Construct one constant-velocity model and one uniformly accelerated model from rest that fit a 5 s trip to 8 m.
**Explain** what more data would resolve.

**Word help:** resolve = separate a vector into components

**Sentence starter:** I predict ... . My model supports this because ... .

**Marzano check - circle:** 1 2 3